Zaheed Bhaiji (he/him)

Vancouver, BC | 604-809-9250 | zaheedbhaiji@gmail.com| LinkedIn| GitHub/Portfolio

EDUCATION

University of British Columbia May 2029

Bachelor of Applied Science – Electrical Engineering

- Co-op Availability: 4, 8 months
- Relevant Courses: CPSC 259 (Data Structures and Algorithms for Electrical Engineers), CPEN 211 (Computing Systems I), ELEC 201 (Circuit Analysis I)

TECHNICAL SKILLS

Lab & Hardware: Oscilloscopes, Function Generators, DMMs, Power Supply, Microcontrollers (ESP32, STM32), FPGA, Sensor Interfacing (UART, I2C), Breadboarding, Circuit Debugging, Soldering
Languages & Simulation: C/C++, Verilog / SystemVerilog, Assembly (RISC-V), Python, MATLAB, FreeRTOS, STM32CubeIDE, Quartus Prime, Questa / ModelSim
CAD & Prototyping: KiCad, Altium Designer, Onshape, SolidWorks, FDM 3D Printing, Waterjet Machining

ENGINEERING DESIGN TEAMS

Uncrewed Aircraft Systems | Vancouver, BC January 2026 - Present
Aircraft Design

- Contributed to building autonomous aerial vehicles for university-level competitions, designing and manufacturing custom airframes from carbon fiber, aluminum and composite materials.
- Collaborated with other sub-teams (electrical, payload) to accommodate and adequately support required electronics and mounted peripherals.

TECHNICAL PROJECTS

Robot Arm (Embedded Systems PCB) | Personal Project July 2026 - Present

- Designed and tested 2-layer PCB in KiCad for a bare-metal ESP32 SoC, with custom boot/flashing circuitry and IP2326 boost charger IC to manage 2S LiPo batteries.
- Used a multi-rail power system with a switching buck converter and LDOs to step voltage down to 5V power and 3.3V logic, calculated trace widths to handle current and prevent overheating.
- Programming FreeRTOS firmware in C with priority levels to read 4 IMUs over I2C, to calculate relative joint angles and drive a 5-DOF arm via PWM signals.

Butterfly Drone Modelling | UAS Design Team April 2026 - June 2026

- Designed and manufactured a custom training drone in OnShape, implementing an open-plane modular chassis to configure different payloads or peripherals.
- Integrated crossbars into frame geometry for torsional rigidity and component mounting space.
- Hand-soldered brushless motors to ESCs, routed signal lines to flight controller pads.

WORK EXPERIENCE

Telus Digital | Langley, BC July 2026 – Present
Online Data Analyst

- Evaluated search queries and machine learning outputs, checking for response relevance and training data accuracy across high-volume datasets.

VOLUNTEER EXPERIENCE

Medical Envoys for Community Health | Langley, BC July 2023 - May 2025
Nonprofit Founder

- Directed a 30+ member student-based organization to spread awareness for local health quality.
- Hosted major community outreach events, raising over $3000 to support community hospitals.